



The Innovator In Pharmaceutical Machinery

Complete Line of Solid Dose Machinery





All Model Product Catalog

Version | V4.2



Complete Line of
Solid Dose Machinery



Tablet Press



Capsule Filler



Tablet Coater



Fluid Bed System



High Shear Mixer

Company Profile

Korean based machinery manufacturer, PTK Co.,Ltd have been busy developing a reputation for robust, technical and cost effective solid dose production machinery for drug or food supplement manufactures.

PTK have utilised the hyper efficient environment of South Korea and have created a complete range of solid dose manufacturing machinery, bringing PTK shoulder to shoulder with other European Brands in terms of output and quality.

PTK is creating the value of pharmaceutical equipment, manufactured by themselves in order to share their vision and satisfy the global community.

The Innovative Leader of Pharmaceutical Machines



‘Good to Great’

Breaking the barrier of current satisfaction, we endeavor to go further. We are dedicating ourselves to reinforce R&D and quality controls as well as adhering to the highest product quality and the most satisfactory service, because we know. ‘Good is enemy of Great!’



Culture and people the bedrock of our success

We are creating the value of pharmaceutical equipments that are manufactured by ourselves in order to share our vision and satisfy global community. Stylish design, robust structures and the state-of-the-art performance of rotary tablet press models, auto capsule filling machines, coating machines and high speed mixers are our key products for the pharmaceutical markets and food industries.

Our Credo

"We demonstrate honesty and sound ethical behavior in all business transactions and personal integrity in all dealings with others."



Harmonized Happiness

We believe in sharing happiness. In the process of growing PTK, we work together with our colleagues, our contractors, our partners and our customers. When we pass through a dark tunnel, we share hope to overcome fear, sadness and frustration. When we get out of the darkness, we share the joy. No matter where you are, this is the journey together in the pursuit of happiness. This is our history we are building together. We are very happy because you are with us together on this journey.



Innovation

Culture and people are the bedrock of our success. Our culture promotes risk taking to find true innovation. We are encouraged to 'fail quickly' and move on. Failure should not be avoided, but accepted as the fair cost while travelling the path to true innovation. We feel motivated and liberated by this supportive approach.



Understanding

Sometimes looking at the problem from a different angle will reveal a solution. Find out how the customer uses our products and their process might reveal a solution or new idea. Think about how that process can be streamlined. Get to know the customer's business. Our customers are all different. Finding the right solution to their problem starts with knowing their business processes, and how they can work best with our systems.



Standardization

We customize to meet the market need. We standardize our procedures to achieve the best result. The critical point is that we are capable of making the balance between customization and standardization. Development, Production, Quality Assurance, Marketing and Sales, share this philosophy. Our united foundation is the implementation of this common standardized platform, from which we are all rewarded with value and efficiency.



Respect

Employees are your internal customers and need a regular dose of appreciation. Thank them and find ways to let them know how important they are. Treat your employees with respect and chances are they will have a higher regard for customers. Appreciation stems from the top. Treating customers and employees well is equally important.



Fun

When appropriate, have fun with the customer. If they're smiling when they hang up, you've done a good job.



Keeping people healthy is the most wonderful mission.
Our approach to this mission is very simple and clear.
Do the right things for our customers.



Who we are

PTK is the innovative leader of pharmaceutical machines whose foundation is secured by the technology of experts and the enthusiasm of the pioneers for further vision.



What we do

Our people are the bedrock of our success. We help them perform to the best of their ability, and make sure they can be proud to work for PTK.



What we follow

We demonstrate honesty and sound ethical behavior in all business transaction and personal integrity in all dealings with others.



We will move toward the best satisfaction.



The beginning of change

Established Palmtech Korea Co., Ltd. in 2005



5 product lines

Providing the best technology and experience



Available anywhere in the world

Fast and satisfactory A/S execution anywhere in the world



Best partner

Continuous effort 365 days a year for customer satisfaction

PTK Global Network

We are providing the quickest and the most satisfactory after sales service.

Asia

- Australia
- Bahrain
- Bangladesh
- China
- India
- Indonesia
- Iran
- Iraq
- Israel
- Japan
- Jordan
- Kazakhstan
- KSA
- Malaysia
- Myanmar
- Oman
- Pakistan
- Philippines
- Singapore
- Syria
- Taiwan
- Uzbekistan

Europe

- Bulgaria
- Cyprus
- Czech
- Finland
- Germany
- Ireland
- Italy
- Lithuania
- Macedonia
- Netherlands
- Poland
- Russia
- Serbia
- Slovakia
- Spain
- Switzerland
- Turkey
- United Kingdom
- Ukraine

America

- Canada
- Chile
- Colombia
- Mexico
- USA
- Venezuela

Africa

- Algeria
- Egypt
- Tunisia





All Model Product Catalog

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Tablet Press

The Frontiers of New Technology!



01



* Specifications are subject to change without notice.

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LABORATORY

PR-LM | Economical

The PTK PR-LM economical model is a compact rotary tablet press for producing small batches in R&D and short run production environments.

- Precisely
- Single Layer
- Core Rod
- 4+4 / 8 / 10 / 12
- HMI



Tooling		B/D	D	B	BB
Number of Stations		8 [4+4]	8	10	12
Max. Main Compression Force	[kN]	40	50	40	40
Max. Pre Compression Force	[kN]	2	3	2	2
Max. Output	[tabs/h]	8,400	16,800	24,000	28,800
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	150			
Turret Speed	[rpm]	5 - 35	5 - 35	5 - 40	5 - 40
Max. Tablet Diameter (Round)	[mm]	16/25	25	16	13
Max. Tablet Dimension (Shape)	[mm]	19/25	25	19	14
Dimension, Machine	[mm]	576 x 851 x 1,460			
Weight, Machine	[kg]	800			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

LABORATORY

PR-LM

The PR-LM series is a compact, ergonomic press designed in accordance with cGMP guidelines and incorporates many of the features found on the larger PTK tablet presses.

- Precisely
- Single Layer
- Core Rod
- 4+4 / 8 / 10 / 12
- HMI



Tooling		B/D	D	B	BB
Number of Stations		8 [4+4]	8	10	12
Max. Main Compression Force	[kN]	40	50	40	40
Max. Pre Compression Force	[kN]	2	3	2	2
Max. Output	[tabs/h]	8,400	16,800	24,000	28,800
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	150			
Turret Speed	[rpm]	5 - 35	5 - 35	5 - 40	5 - 40
Max. Tablet Diameter (Round)	[mm]	16/25	25	16	13
Max. Tablet Dimension (Shape)	[mm]	19/25	25	19	14
Dimension, Machine	[mm]	575 x 570 x 1,460			
Dimension, Controls		500 x 500 x 1,182			
Weight, Machine	[kg]	700			
Weight, Controls	[kg]	90			
Power Supply Data		220/380/400/ 415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			



Video

LABORATORY

PR-LD

The PR-LD series is a tablet press capable of producing single or double layer tablets. It is the ideal system for R&D.

- Precisely
- Single Layer
- Double Layer
- Core Rod
- Turret Change
- 11+11 / 22 / 27 / 32 / 34
- HMI



Video

Tooling		B/D	D	B	BB	BBS
Punch Type / Die Type		22 [11+11]	22	27	32	34
Double Layer Tablet						
Max. 1st Compression Force	[kN]	20				
Upper Punch Penetration Depth	[mm]	1-4[Main], 1-8[Pre 1st], 1-4[Pre 2nd]				
Turret Speed	[rpm]	5 - 30				
Max. Output	[tabs/h]	19,800	39,600	48,600	57,600	61,200
Single Layer Tablet						
Max. Main Compression Force	[kN]	70				
Max. Pre Compression Force	[kN]	20				
Turret Speed	[rpm]	5 - 70				
Max. Output	[tabs/h]	46,200	92,400	113,400	134,400	142,800
Max. Filling Depth	[mm]	18				
Max. Tablet Thickness	[mm]	8.5				
Upper Punch Insertion	[mm]	1 - 4				
Pitch Circle Diameter	[mm]	315				
Max. Tablet Diameter (Round)	[mm]	16/25	25	16	13	11
Max. Tablet Dimension (Shape)	[mm]	19/25	25	19	14	12
Dimension, Machine	[mm]	1,000 x 850 x 1,810				
Dimension, Controls	[mm]	730 x 630 x 1,700				
Weight, Machine	[kg]	2,700				
Weight, Controls	[kg]	120				
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz				
Control Power		24Vdc				

LABORATORY

PR-LT

The PTK PR-LT model tablet press is capable of producing multi-layered tablets. [single, double and triple layer] The PR-LT is ideal for laboratory use. The sequence feeder system makes it possible to produce sample tablets with minimal powder.

- Precisely
- Single Layer
- Double Layer
- Triple Layer
- Core(Tab-In-Tab)
- Core Rod
- Turret Change
- 8+8 / 20 / 24 / 29 / 31
- HMI



Video

Tooling		B/D	D	B	BB	BBS
Number of Stations		16 [8+8]	20	24	29	31
Max. Main Compression Force	[kN]	80	100	80	80	80
Max. Pre Compression Force	[kN]	80	100	80	80	80
Max. Filling Depth	[mm]	18				
Max. Tablet Thickness	[mm]	8.5				
Upper Punch Insertion	[mm]	1 - 6				
Pitch Circle Diameter	[mm]	285				
Turret Speed	[sec/rev]	2 - 10				
Tablet Output		1 tablet / 3 rotation				
Max. Tablet Diameter (Round)	[mm]	16/25	25	16	13	11
Max. Tablet Dimension (Shape)	[mm]	19/25	25	19	14	12
Dimension, Machine	[mm]	1,700 x 1,138 x 1,870 [Control Cabinet Included]				
Weight, Machine	[kg]	2,500				
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz				
Control Power		24Vdc				

PR-LT

All-In-One R&D Function

Every Type of Tablet Research Available



The PTK PR-LT Series is capable of producing single, double and triple layer tablets. With a quick part change, it also produces core rod tablets and core tablets [tab-in-tab]. The PR-LT Series is the only tablet press for laboratory use which can produce MUPS [Multiple Unit Pellet System] tablets. Only a change of powder hopper is needed.

The strength of the PR-LT series is that it collects multiple data points, based on an individual station, recorded on a production size turret. Therefore, testing on this machine bridges the gap of upscale uncertainties and gives true life results on 1 single compression or continuously.

The PR-LT Series provides the best solution for laboratory unit as all kind of types of tablet production is available. And PR-LT is designed to be available with scaling up to pilot with lower cost.



Sequence Feeder



Core Tablet Feeder



MUPS



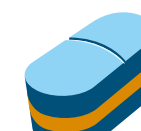
Mechanical Feeder



Single Layer



Double Layer



Triple Layer



Core (Tab-In-Tab)



Core Rod



MUPS

Simple and Special One Stop Solution



B&D Turret

Uses standard D, B and BB tooling types.

Or with the option of a B&D Combination Turret. With this option a cam change is enough to make a single turret compatible with B and D Tooling.



Wide Range of Filling Cams

With a range of various filling cams [0.5 to 18 mm].

It is possible to meet the customer's various needs.

This critical point reduces powder loss, and replicates production models.



Direct Drive Motor

PTK's Torque Drive Motor replaces the conventional belt driven motor and gearbox assembly. Therefore reducing heat and noise. Heat is reduced further by selecting the water-cooled motor with an intermittent water supply.

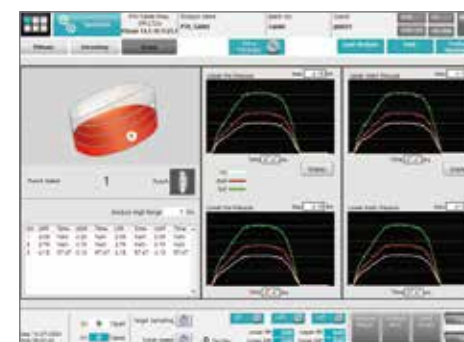
For optimum cool running use PTK's PISCon software to set the Interval time & supply time at the touch screen control panel.



Sequence Feeder

The Sequence Feeder is an ingenious feeding system allowing single dose feeding for 1,2, or 3 layer tablets on a compact R&D Tablet Press.

This Servo Motor driven device accurately doses each tablet and subsequent layers individually, therefore using the minimum powder to create a single or small quantity of tablets.



Dwell Time Monitoring

Dwell time plays an important role in tablet manufacturing process. It occurs when compression rollers are in contact with the punch head flat.

At the PTK HMI system, 1st layer, 2nd layer and 3rd layer dwell time expressed in milliseconds with Sequence Feeder.



PILOT

PR-1000

The PTK PR-1000 series is an economical, compact, single-sided rotary tablet press with pre-compression designed in accordance with cGMP guidelines.

-  Precisely
-  Single Layer
-  Core Rod
-  16 / 19 / 23 / 25
-  HMI



Tooling		D	B	BB	BBS
Number of Stations		16	19	23	25
Max. Main Compression Force	[kN]	70	60	60	60
Max. Pre Compression Force	[kN]	30	20	20	20
Max. Output	[tabs/h]	96,000	136,800	165,600	180,000
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	225			
Turret Speed	[rpm]	15 - 100	15 - 120	15 - 120	15 - 120
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	900 x 750 x 1,740			
Dimension, Controls	[mm]	650 x 600 x 1,502			
Weight, Machine	[kg]	1,900			
Weight, Controls	[kg]	120			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PRODUCTION

PR-1500

The PTK PR-1500 series is a modern, single-sided rotary tablet press designed especially for unattended operation in cGMP environments.

-  Precisely
-  Single Layer
-  Core Rod
-  Turret Change
-  20 / 24 / 29 / 31
-  HMI



Tooling		D	B	BB	BBS
Number of Stations		20	24	29	31
Max. Main Compression Force	[kN]	100	80	80	80
Max. Pre Compression Force	[kN]	100	80	80	80
Max. Output	[tabs/h]	120,000	172,800	208,800	223,200
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	285			
Turret Speed	[rpm]	15 - 100	15 - 120	15 - 120	15 - 120
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,000 x 806 x 1,916			
Dimension, Controls	[mm]	650 x 600 x 1,502			
Weight, Machine	[kg]	2,500			
Weight, Controls	[kg]	120			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PRODUCTION

PR-2000

The PTK PR-2000 series is a state-of-the-art computer controlled, single-sided rotary tablet press designed especially for unattended operation in cGMP environments.

- Precisely
- Single Layer
- Core Rod
- Turret Change
- 29 / 36 / 43 / 47
- HMI



Video

Tooling		D	B	BB	BBS
Number of Stations		29	36	43	47
Max. Main Compression Force	[kN]	100			
Max. Pre Compression Force	[kN]	100			
Max. Output	[tabs/h]	174,000	216,000	258,000	282,000
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	415			
Turret Speed	[rpm]	15 - 100			
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,190 x 985 x 1,880			
Dimension, Controls	[mm]	650 x 600 x 1,502			
Weight, Machine	[kg]	3,300			
Weight, Controls	[kg]	120			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PRODUCTION

PR-2000F

The PTK PR-2000F series is a state-of-the-art computer controlled, single-sided rotary tablet press designed especially for unattended operation in cGMP environments.

- Precisely
- Single Layer
- Core Rod
- Turret Change
- 29 / 36 / 43 / 47
- HMI



Video

Tooling		D	B	BB	BBS
Number of Stations		29	36	43	47
Max. Main Compression Force	[kN]	100			
Max. Pre Compression Force	[kN]	100			
Max. Output	[tabs/h]	174,000	216,000	258,000	282,000
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	415			
Turret Speed	[rpm]	15 - 100			
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,190 x 985 x 1,880			
Dimension, Controls	[mm]	650 x 600 x 1,502			
Weight, Machine	[kg]	3,300			
Weight, Controls	[kg]	120			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PRODUCTION

PR-2100

The PTK PR-2100 series featuring WIP [Wash-In-Place], is designed to protect the operators from hazardous or toxic products.

- Precisely
- Single Layer
- Core Rod
- Turret Change
- WIP
- Containment
- 35 / 43 / 51 / 55
- HMI



Tooling		D	B	BB	BBS
Number of Stations		35	43	51	55
Max. Main Compression Force	[kN]	100			
Max. Pre Compression Force	[kN]	100			
Max. Output	[tabs/h]	168,000	258,000	306,000	330,000
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	495			
Turret Speed	[rpm]	15 - 80	15 - 100	15 - 100	15 -1 00
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,236 x 1,119 x 1,850			
Dimension, Controls	[mm]	650 x 600 x 1,502			
Weight, Machine	[kg]	4,500			
Weight, Controls	[kg]	120			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PRODUCTION

PR-3000

The PTK PR-3000 series is a double-sided press capable of efficient, high speed production of single layer or double layer tablets.

- Precisely
- Single Layer
- Double Layer
- Core Rod
- Turret Change
- 43 / 52 / 63 / 68
- HMI



Video

Tooling		D	B	BB	BBS
Number of Stations		43	52	63	68
Max. Main Compression Force	[kN]	100			
Max. Pre Compression Force	[kN]	100			
Max. Output, Single Layer	[tabs/h]	309,600	436,800	529,200	571,200
Max. Output, Double Layer	[tabs/h]	103,200	156,000	189,000	204,000
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	Pre [1-5] Main [1-8]			
Pitch Circle Diameter	[mm]	615			
Turret Speed, Single Layer	[rpm]	5 - 60	5 - 70	5 - 70	5 - 70
Turret Speed, Double Layer	[rpm]	5 - 40	5 - 50	5 - 50	5 - 50
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,550 x 1,200 x 1,990			
Dimension, Controls	[mm]	730 x 630 x 1,602			
Weight, Machine	[kg]	5,700			
Weight, Controls	[kg]	140			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PRODUCTION

PR-3500

The PTK PR-3500 series is a triple compaction tablet press capable of producing single, double and triple layer tablets. PR-3500 is designed for today's current cGMP pharmaceutical manufacturing environments.

- Precisely
- Single Layer
- Double Layer
- Triple Layer
- Core(tab-in-tab)
- Core Rod
- 43 / 52 / 63 / 68
- HMI



Tooling		D	B	BB	BBS
Number of Stations		43	52	63	68
Max. Main Compression Force	[kN]	100			
Max. Pre Compression Force	[kN]	30			
Max. Output, Single Layer	[tabs/h]	90,300	124,800	151,200	163,200
Max. Output, Double Layer	[tabs/h]	77,400	109,200	132,300	142,800
Max. Output, Triple Layer	[tabs/h]	64,500	93,600	113,400	122,400
Max. Filling Depth	[mm]	2 - 6 [1st], 3.5 - 10 [2nd], 5 - 14 [3rd]			
Max. Tablet Thickness	[mm]	9			
Upper Punch Insertion	[mm]	2.5 - 6 [1st, 2nd], 1 - 4 [3rd]			
Pitch Circle Diameter	[mm]	615			
Turret Speed, Single Layer	[rpm]	5 - 35	5 - 40	5 - 40	5 - 40
Turret Speed, Double Layer	[rpm]	5 - 30	5 - 35	5 - 35	5 - 35
Turret Speed, Triple Layer	[rpm]	5 - 20	5 - 30	5 - 30	5 - 30
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,656 x 1,435 x 1,875			
Dimension, Controls	[mm]	730 x 630 x 1,602			
Weight, Machine	[kg]	5,000			
Weight, Controls	[kg]	140			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PRODUCTION

PR-4000

The PTK's highest-capacity double-sided tablet press model, PR-4000 guarantees a plentiful output and is designed to produce tablets in an efficient double-sided configuration.

- Precisely
- Single Layer
- Double Layer
- Core Rod
- Turret Change
- 60 / 73 / 87 / 93
- HMI



Video

Tooling		D	B	BB	BBS
Number of Stations		60	73	87	93
Max. Main Compression Force	[kN]	100			
Max. Pre Compression Force	[kN]	100			
Max. Output, Single Layer	[tabs/h]	468,000	657,000	783,000	837,000
Max. Output, Double Layer	[tabs/h]	108,000	175,200	208,800	223,200
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	Pre [1-5] Main [1-8]			
Pitch Circle Diameter	[mm]	850			
Turret Speed, Single Layer	[rpm]	15 - 65	15 - 75	15 - 75	15 - 75
Turret Speed, Double Layer	[rpm]	5 - 30	5 - 40	5 - 40	5 - 40
Max. Tablet Diameter (Round)	[mm]	25	16	13	11
Max. Tablet Diameter (Shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,626 x 1,382 x 2,130			
Dimension, Controls	[mm]	730 x 630 x 1,602			
Weight, Machine	[kg]	6,900			
Weight, Controls	[kg]	300			
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

PR-4000 Series for High Capacity Tablet Production



Reduce Tablet Weight Variation with
Elongated Cam Track Design.

Extended Feeder Length for Accurate
Dosing at High Speed.

- Double Sided Single layer & Single-Sided
Double Layer Tablet Production
- 60 / 73 / 87 / 93 stations
- Inflatable Door Seal for Complete Dust
Containment and Noise Reduction
- Changeable Turret Option for Rapid
Machine Setup
- 21 CFR Part 11 conformity

High Speed and High Pressure Technology



Down Force Ram

Powerful air cylinders are equipped on the top of the machine to secure the turret for stable operation with high compression force.



Soft Start

To combat punch overloading when accelerating or decelerating to and from high speed, the machine rapidly relieves pressure until full operating speed is achieved.

The machine will only accept good tablets and controls variations on machine start and stop.

This function is selectable so only used when necessary.

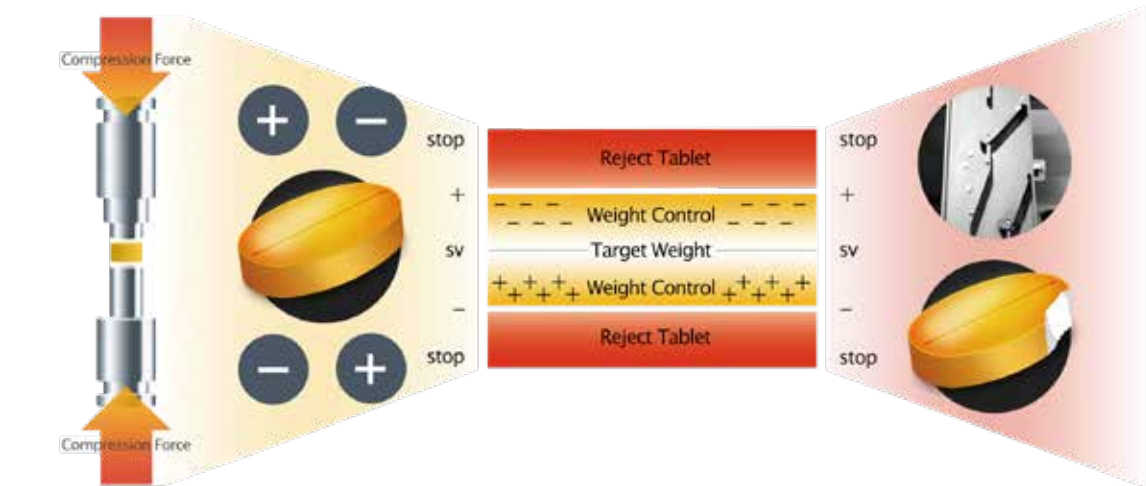


Precision Cam Track

The upper and lower rail manufactured with a specialized high precision tool makes the turret possible to run at the highest speed and reduce the noises.

In-House Program with 21 CFR Part 11 Compliance

Weight Control Function



Tablet Rejection Function



Automatic Weight Regulation

Automatic Weight Regulation monitors and regulates tablet weights within a specified range. The 3 channel system manages Good, Reject and Sample tablets. Automated with the weight regulation system to ensure 100% of tablets entering the good stream are monitored.



In-House Program

With our in-house Programmed software, PTK provides maximized customization of the software.

This makes for a fast, light and stable program with intuitive interface. Opened protocol means it is possible to connect OEE system which shows real time productivity, availability and quality.



Punch Tightness Monitoring

The punch tightness of the upper and lower punches can be monitored and any excessive pressure [tightness] can be detected.

If the movement of the upper and lower punch in the upper disk hole is not smooth, the load-cell on this rail detects it and stops the machine. The tightness limit is to be set in PISCon. If the tightness level goes beyond the limit, the machine will activate an alarm.



Data Collection

The control systems available are dependent upon the customer's specified level of system control and the sophistication level of data collection and analysis to be performed by the control system. An unlimited number of data storage also guarantees maximized traceability.

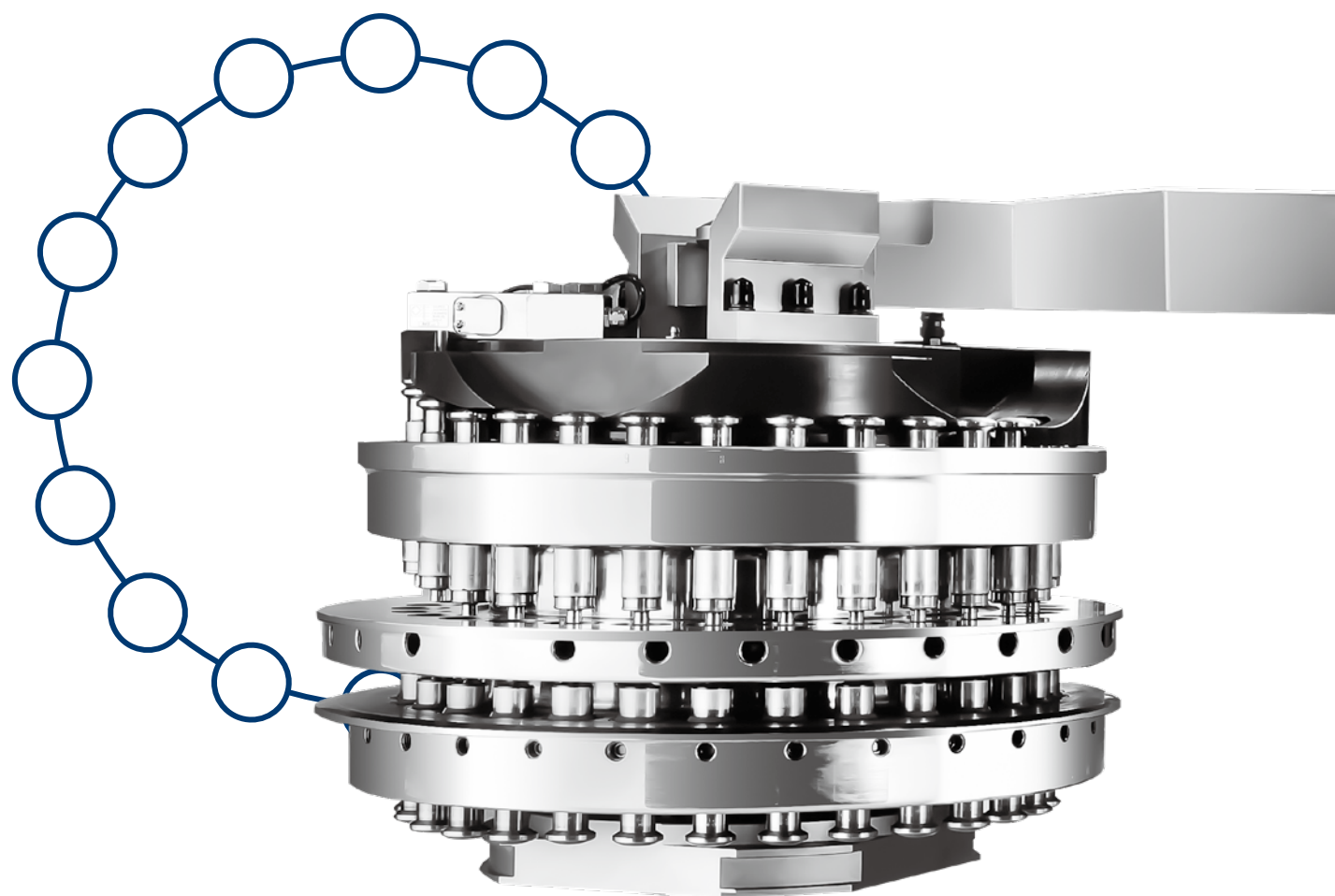


Dwell Time Monitoring

Dwell time takes place when the compression rollers make contact with the punch head flat. It greatly impacts the overall quality of many tablet products.

During the compression, the dwell time displayed on the HMI screen with Dwell Time Report.

Removable Turret System Benefits



- Faster product to product change-over
- Easier production planning and preparation as machine turrets can be made ready in advance
- The ability to clean to a higher standard and offline
- Configure the machine to maximize output through optimum tooling to product size selection
- Keep the initial investment down by buying the option, and add different turret sizes at a later date.



Product Change-Over Now Even Faster



30 Minute Change-Over

PTK tablet presses now contain both upper and lower punches on turret change, driving down turret change times even lower with complete machine change over achievable in 30 minutes.

Change over times can be optimized using additional change parts with spare turrets pre-tooled to include the lower tooling. Easy to remove lower cams, quick release dust extraction and removable turret, means access for cleaning is made very easy.



Curvic Coupler

A hardened precision coupling is used to connect the machine to each turret. Ensures repeatable turret re-location to hundredths of a millimetre.

Eliminates wear and vibration due to direct opposing lock of turret to machine. Allows complete accuracy of the turret's rotary position at all times.



Servo Motor

Position control with high accuracy is possible by the use of servo motors. These control the upper and lower roller, dosing cam and feeder drive. With a displayed range of 100th mm, the actual position range is 1000th mm.

Encoder feedback gives position information and allows for perfect changeover repeatability.

Maximized Customization with Multi-Layer Tablet



Scraping

Before 2nd level feeding, a precision PEEK scraper is equipped to avoid powder contamination. With air purge and suction capability it prevents powder contamination effectively.



First Layer Sampling

When the press is configured for double layered tablets the machine is capable of setting and monitoring the first layer weight by ingeniously discharging the initial layer only.

This is done by means of a pneumatic cylinder lifting the rail forcing first layer ejection. The second layer is then recognized as being underweight and ejected in due course.



Double Layer & Single Layer

The press can be configured with one or two output chutes.

Conventional tablets can be manufactured with double sided pre and main compression for highest outputs, or for double layer tablets by single tablet output.

The change-over process is easily possible by the user with the most basic of training.



Feeder-Segment

The Feeder segment is a wear part, making it feasible to maintain the powder retaining aspect of the feeder with hard wearing materials to maintain high batch yields.

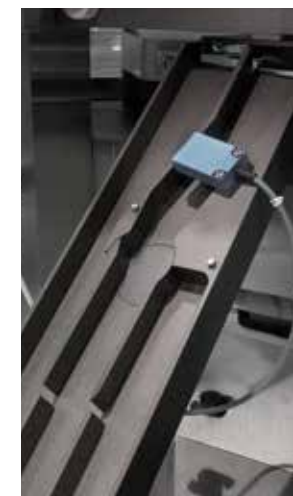
The selected materials have natural lubricity, therefore protecting the turret surface in case of accidental contact.

Optional Items



Initial Reject & Sampling Gates

Features automatic timer settings for opening intervals and duration to reject tablets on start-up and shut-down and for automatic tablet sampling in production.



Chute Sensor

Detects jams and blockages in the discharge chute. Programmable alarm feature alerts operators and shuts down the press.



Open Feeder

For good flowing materials or applications where a mechanical feeder is not suitable i.e. segregation problems, special matrices or short runs.



IFH

It supplies fixed amount of powder to the powder feed tube by rotating of the IFH roller. It is useful when the powder has poor flow.



Direct Drive Motor

Replace conventional belt driven motor & reducer to improve torque. With the water-cooled motor system, heat emission can be reduced.



Turret Table

Allows for the safe handling and easy transport of removable turrets for installation, removal or maintenance.

Optional Items



Powder Loader [Built In Type]

The built-in powder loader, operates in the same way as the standard powder loader, by using the Vacuum transfer method. This option features installation of the control valving inside the upper structure of the tablet press frame.



Powder Loader

Vacuum transfer system supplies the powder automatically to the main hopper on the tablet press from floor level.



Dust Collector

This device extracts dust from the dust extraction system at the machine's compression zone during the operation.



Tablet Deduster

Uphill deduster removes dust adhered to the surface of tablets by the Vibrator. It is connected to the machine for easy control on the screen and can be connected Dust Collector to get the removed dust from the deduster.



Tablet Deduster Combined with Metal Detector

Uphill type tablet deduster and Metal detector combined together with one frame. It can be connected to the tablet press machine so the operation can be linked with tablet press.



Metal Detector

The system is built with a gas strut stand frame offering exceptional manoeuvrability making it convenient for use with tablet presses and capsule machines. Head size 95x38mm with lift-flap rejection system will be provided.

Optional Items



UTS [Sampling Checker]

The flexible and modular Kraemer Tablet Testing System, UTS automatically measures tablet weight, thickness, diameter and hardness with IPC control with the machine. Also, additional feedback function is available.



Magnesium Spray Device

The Magnesium spray device is useful for sticky products, particularly when this causes weight variation, or high ejection pressure. It sprays the magnesium to the punch tips.



Calibration Kit

The calibration kit can be used periodically to check the correct operation of the load cell. The reference punches, load cell and indicator with 3rd part calibration will be provided together.



Capsule Filler

The Frontiers of New Technology!



02



* Specifications are subject to change without notice.

PF-1000	44
PF-2000	45
PF-2000F	46
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Optional Items	52
PCW-Series	54

PILOT

PF-1000

The PTK PF-1000 series, automatic capsuling machines are designed to automatically fill powders, pellets or tablets into two-piece capsules.

-  Powder Filling
-  Pellet Filling
-  Tablet Filling
-  HMI



Video

Number of Holes / Segment	1	2	3	4	5	6
Max. Speed, Powder [rpm]	135					
Max. Speed, Pellet [rpm]	80					
Max. Output, Powder [caps/h]	8,100	16,200	24,300	32,400	40,500	48,600
Max. Output, Pellet [caps/h]	4,800	9,600	14,400	19,200	24,000	28,800
Power of Main Motor [kW]	1.5					
Air Consumption [liter/min]	650					
Compressed Air [bar]	6					
Dimension, Machine [mm]	1,000 x 841 x 1,760					
Dimension, Controls [mm]	650 x 600 x 1,502					
Weight, Machine [kg]	1,420					
Weight, Controls [kg]	120					
Capsule Size	#000, #00, #0, #1, #2, #3, #4					
Power Supply Data	220/380/400/415/440/460Vac, 3Ph, 50/60Hz					
Control Power	24Vdc					

PRODUCTION

PF-2000

The PTK PF-2000 series is an ideal solution for environments with moderate production requirements.

-  Powder Filling
-  Pellet Filling
-  Tablet Filling
-  HMI



Number of Holes / Segment	10	12	14
Max. Speed, Powder [rpm]	125		
Max. Speed, Pellet [rpm]	75		
Max. Output, Powder [caps/h]	75,000	90,000	105,000
Max. Output, Pellet [caps/h]	45,000	54,000	63,000
Power of Main Motor [kW]	2.2		
Air Consumption [liter/min]	850		
Compressed Air [bar]	6		
Dimension, Machine [mm]	1,230 x 1,230 x 1,716		
Dimension, Controls [mm]	650 x 600 x 1,502		
Weight, Machine [kg]	2,200		
Weight, Controls [kg]	120		
Capsule Size	#000, #00, #0, #1, #2, #3, #4		
Power Supply Data	220/380/400/415/440/460Vac, 3Ph, 50/60Hz		
Control Power	24Vdc		

PRODUCTION

PF-2000F

The PTK PF-2000F series is an ideal solution for environments with moderate production requirements.

-  Powder Filling
-  Pellet Filling
-  Tablet Filling
-  HMI



Video

Number of Holes / Segment		10	12	14
Max. Speed, Powder	[rpm]	125		
Max. Speed, Pellet	[rpm]	75		
Max. Output, Powder	[caps/h]	75,000	90,000	105,000
Max. Output, Pellet	[caps/h]	45,000	54,000	63,000
Power of Main Motor	[kW]	2.2		
Air Consumption	[liter/min]	850		
Compressed Air	[bar]	6		
Dimension, Machine	[mm]	1,250 x 1,250 x 1,950		
Dimension, Controls	[mm]	650 x 600 x 1,502		
Weight, Machine	[kg]	2,080		
Weight, Controls	[kg]	120		
Capsule Size		#000, #00, #0, #1, #2, #3, #4		
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz		
Control Power		24Vdc		

PRODUCTION

PF-3000

The PTK PF-3000 series is especially suited for production environments where maximum throughput and high capacity is required.

-  Powder Filling
-  Pellet Filling
-  Tablet Filling
-  HMI



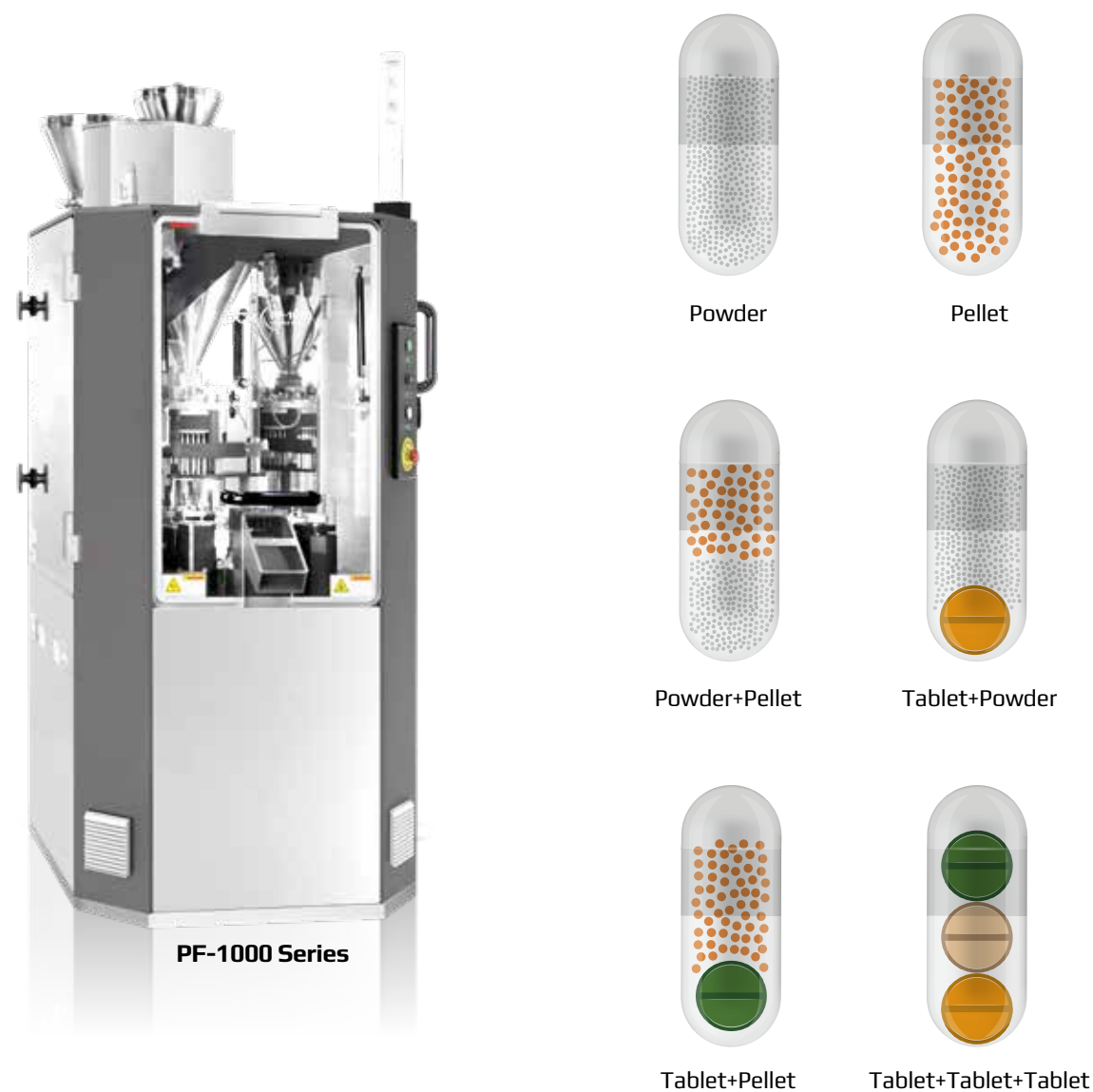
Number of Holes / Segment		16	18	20
Max. Speed, Powder	[rpm]	125		
Max. Speed, Pellet	[rpm]	75		
Max. Output, Powder	[caps/h]	120,000	135,000	150,000
Max. Output, Pellet	[caps/h]	72,000	81,000	90,000
Power of Main Motor	[kW]	3.7		
Air Consumption	[liter/min]	1,050		
Compressed Air	[bar]	6		
Dimension, Machine	[mm]	1,450 x 1,300 x 1,810		
Dimension, Controls	[mm]	650 x 600 x 1,502		
Weight, Machine	[kg]	2,800		
Weight, Controls	[kg]	120		
Capsule Size		#000, #00, #0, #1, #2, #3, #4		
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz		
Control Power		24Vdc		

Available of Various Filling Combinations.



Various Filling Combinations

With Powder, Pellet and Tablet Filling station, it is capable to have various combination in a capsule upon the requirement.



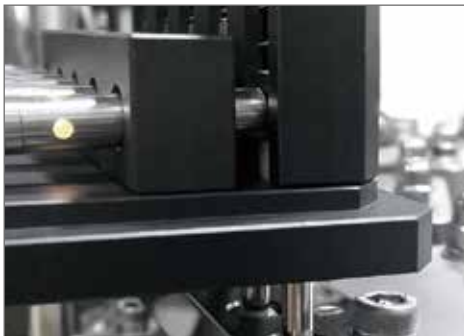
Capsule Filler with Control Cabinet

Integrated capsule filler and control cabinet

Significantly minimizing installation space through a unified design.



Our Encapsulation Technology

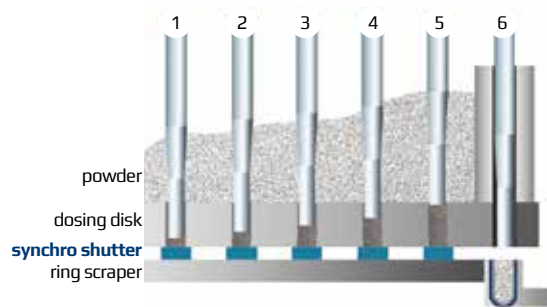


Auto De-Jamming Device

If the sensor cannot recognize the capsule after the capsule orientation and before the filling station, de jamming device blows the air to removed jammed capsules at the feeder automatically.

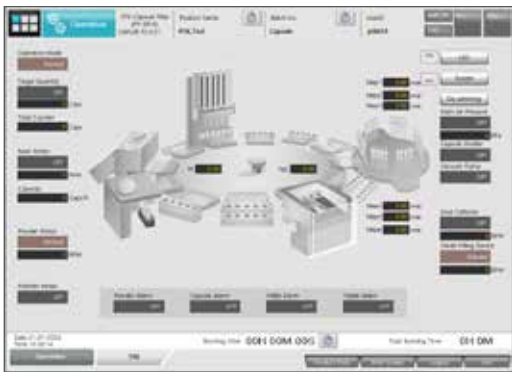
Synchro Shutter System

With a revolutionary dosing disk, the synchro shutter system minimizes the gap between the shutter and the ring scraper. This minimizes powder loss through powder burst at the initial compaction causing sifting, holding true perfect accuracy right through each tamping stage. It provides an optimized solution for hard-to-press, or sticky product by containing the dose, eliminating friction on the product and maintaining clean component parts with in the machine.



In-house Program with 21 CFR Part 11 Compliance

The operation of the PF Series is a fine example of user-friendly design. The software ensures the optimum capsule quality with its production data management and automation control. The product data settings can be stored and re-called to aid uniform production.



Overview



Setting



Recipe



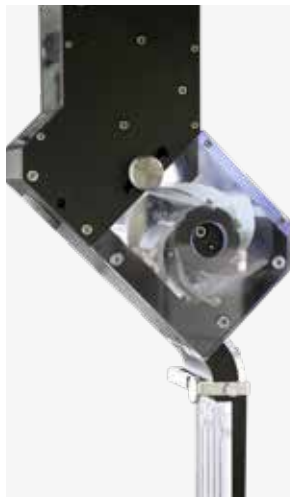
Safety

Optional Items



Pellet Filling Device

The pellet device is for filling pellets into hard gelatin capsules. It is designed to minimize weight deviation.



Tablet Filling Device

The tablet filling device uses innovative orientation and separation technology to accurately dispense tablets into Hard Gelatin Capsules. Various combinations of tablet, powder, pellet will be available with this device, too.



Dust Collector

Stainless steel dust collector connected with machine and extracts dust from the powder tamping & etc during the operation.



Change Parts

The machine can be configured for all capsule sizes. So, additional set of change parts including dosing disk are available depending on the requirement.



Capsule Loader

By the air from the blower transfer the empty capsules to the machine capsule hopper automatically. The speed of the blower can be controlled by the speed controller.



Powder Loader

Vacuum transfer system supplies the powder automatically to the main powder hopper of the machine.

Optional Items



Capsule Polisher

The dust on the produced capsules can be removed by the screw brush inside the capsule polisher.



Capsule Polisher With Sorter

Empty or light weight produced capsules can be rejected to the Reject Bin with sorter option.



Capsule Polisher Combined with Metal Detector

Removed external dust on the finished capsules while them transferring upward by the screw bursh and passing the metal checker with combination design. WIP option is available at this combination model.

PRODUCTION

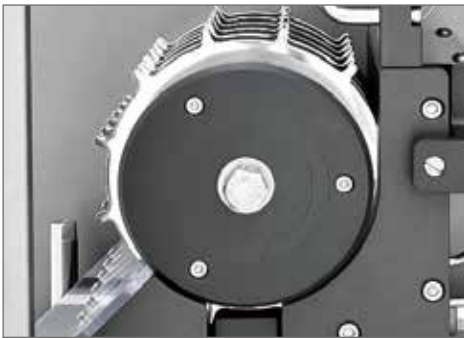
PCW-Series

PTK Capsule Weighing System, 'WEISYS' capable of rejecting the capsules which beyond the target weight automatically by the high-precision cells at the machine. HMI System with 21 CFR Part 11 Conformity is configured for user's convenience. Meet and experience PTK Capsule Weighing System with various functions such as Initial & Auto rejection, jammed capsule detection & etc.



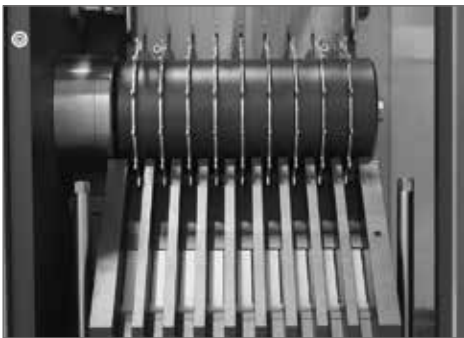
Model		PCW-6	PCW-9	PCW-12	PCW-18
Max Capsule Output	[cap/hr]	60,480	90,720	120,960	181,440
Capsule Size		#000,#00, #0, #1, #2, #3, #4			
Ambient Temp		15°C to 25°C			
Weight Accuracy	[mg]	±2			
Measurement Range	[mg]	0~1,800			
Control System		IPC with Window 10			
Compressed Air	[bar]	6			
Dimension	[mm]	900 x 900 x 1,552		1,100 x 900 x 1,600	1,400 x 900 x 1,600
Weight	[kg]	650	700	850	1,050
Power supply data		220Vac, 1ph, 50/60Hz			
Control Power		24Vdc			

Our Capsule Weighing System Technology



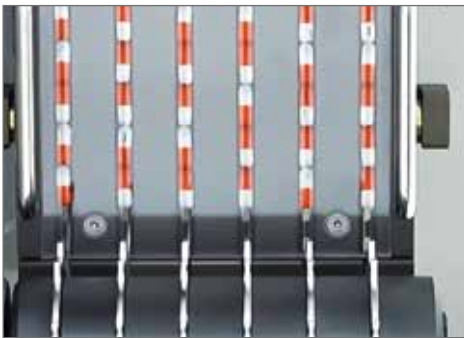
Star Wheel & Servo Motor

Star wheel rotated by the servo motor so each capsules can be located on the weight cell correctly.



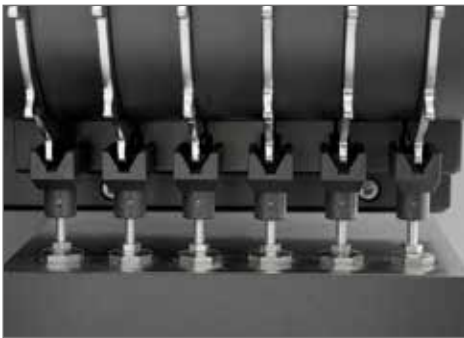
Auto Rejection

The capsules which beyond the target weight will be rejected automatically to the bin at the discharge chute.



Auto De-jamming System

If there's jammed capsule in the feeder, de-jamming system can be functioned and remove the jammed capsules. Manual or Auto mode are available and it can be set on the HMI screen easily.



Capsule Weighing Cell

Weighing cell with air purging system to remove dust on the capsule weight measuring platform. Error rate of load cell can be minimized.



Tablet Coater

The Frontiers of New Technology!



03



* Specifications are subject to change without notice.

PC-L	58
PC-C 620	59
PC Series Chamber Type	60
Optional Items	66

LABORATORY
PC-L

The PTK tablet coating machine model, PC-L series is a self-contained, laboratory scale tablet coater featuring a built-in air handling system for inlet air filtration, heating and dust collection.

- Film Coating
- Pan Changeable
- HMI



Video

Model	PC-L 300					
Drum Diameter	[mm]	300 (S)	300	320	430	450
Drum Opening Diameter	[mm]	170	170	170	230	230
Drum Working Volume	[ℓ]	0.2 - 0.7	0.4 - 1.1	0.8 - 2.2	2 - 4.5	4 - 5.5
Output Capacity *	[kg]	0.14 - 0.5	0.3 - 0.8	0.6 - 1.5	1.4 - 3.2	2.8 - 3.9
Pan Speed	[rpm]	0 ~ 40				
Spray Gun		1				
Weight, Machine	[kg]	710				
Dimension, Machine	[mm]	1,200 x 1,230 x 1,980				
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz				
Control Power		24Vdc				

* Specific Gravity 0.7

PILOT
PC-C 620

PTK's tablet coaters precisely apply a spray of a liquid tablet coating system onto the tablets. A directed flow of drying air is drawn through the bed of tablets as they are gently agitated inside the rotating pan.

- Film Coating
- Film & Sugar Coating
- Pan Changeable
- HMI



Video

Model		PC-C 620		
Drum Diameter	[mm]	500	600	700
Drum Opening Diameter	[mm]	300		
Drum Working Volume	[ℓ]	4.0 - 7.0	9.0 - 16.0	17.0 - 28.0
Output Capacity *	[kg]	2.8 - 4.9	6.3 - 11.2	11.9 - 19.6
Pan Speed	[rpm]	0 ~ 20		
Spray Gun		1		
Drive Power	[kW]	0.75		
Weight, Machine	[kg]	500	600	650
Weight, Controls	[kg]	90		
Dimension, Machine	[mm]	1,050 x 1,463 x 1,507		
Dimension, Controls	[mm]	661 x 677 x 1,278		
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz		
Control Power		24Vdc		

* Specific Gravity 0.7

PRODUCTION

PC Series | Chamber Type



The coating pan with inlet & outlet duct of these machines are designed with focused airflow to provide more efficient drying, minimizing air usage. Further improved machine structure applied to Chamber type to increase the heating efficiency and more easy cleaning.

- Film & Sugar Coating
- HMI



Model	PC-820	PC-1020	PC-1320	PC-1520	PC-1520i	PC-1720	PC-1720i	PC-1720t
Drum Diameter [mm]	820	1,050	1,300	1,500	1,500	1,700	1,700	1,700
Drum Opening Diameter [mm]	380	420	500	578	578	600	600	600
Drum Working Volume [ℓ]	24 - 45	57 - 100	100 - 180	160 - 280	200 - 362	250 - 440	340 - 600	450 - 800
Output Capacity * [kg]	17 - 32	40 - 70	70 - 126	112 - 196	140 - 253	175 - 308	238 - 420	315 - 560
Pan Speed [rpm]	0 ~ 16							0 ~ 12
Spray Gun	2	2	2	3	3	4	4	5
Drive Power [kW]	2.2	2.2	3.7	3.7	3.7	7.5	7.5	7.5
Weight, Machine [kg]	1,450	1,650	1,850	2,000	2,150	2,250	2,400	3,000
Weight, Controls [kg]	90							
Dimension, Machine [mm]	1,200 x 1,610 x 1,682	1,460 x 1,585 x 1,757	1,720 x 1,705 x 2,005	2,050 x 1,818 x 2,244	2,050 x 2,038 x 2,244	2,254 x 2,008 x 2,392	2,254 x 2,258 x 2,392	2,254 x 2,648 x 2,392
Dimension, Controls [mm]	661 x 677 x 1,278							
Power Supply Data	220/380/400/415/440/460Vac, 3Ph, 50/60Hz							
Control Power	24Vdc							

* Specific Gravity 0.7

Tablet Coater Chamber Type

Meet and experience the Human-friendly coating system which shows the new paradigm of coating technology.



In-house Program with 21 CFR Part 11 Compliance

The coating process can be easily automated and configured for the optimal production environment, which guarantees exceptional product quality.

The user-friendly touch screen controls make for a highly efficient production environment. Batch report and data analysis functions are available with Tablet Coater, High Shear Mixer and Fluid Bed System.



Liquid Volume Prediction System

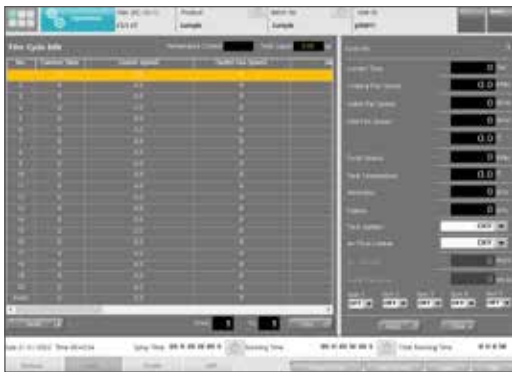
The Liquid Volume Prediction System [LVPS] calculates the Coating Solution supply rate, and therefore also, the total usage per batch. This is set based on an accumulation of initial sample data based on flow rates for specific sample speeds.

The system therefore accurately calculates the total liquid for one whole batch to enable consistent planned consumption of liquid solution [minimized loss] and process repeatability. The HMI option includes this system without any extra cost.

Now, the key functions of a flow meter are all available within PTK's HMI in-house software. So free from any cleaning and maintenance.



Overview



Cycle



WIP



Batch Report

Measurement of the Flow Rate at Any Programmed Speed



Data Input



Display of Predicted Volume for Each Cycle

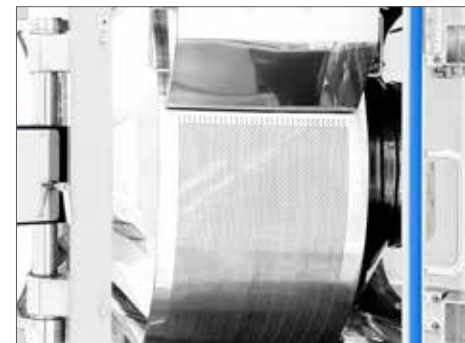


Our Coating Technology



Improved WIP System

The coating pan & duct in the machine can be cleaned automatically with this system. WIP nozzles are installed inside & outside of the coating pan for further cleaning. Through the WIP Rack, detergent, hot water, cold water and purified water can be supplied to the machine and auto cycle by the software is available on the screen.



Perforated Coating Pan

Baffles are equipped in the coating pan for the fast mixing of the tablets to have high quality of coating performance. There's perforation on the coating pan. For production model, 2.5X10mm or 3mm is available.



Inflatable Seal

Coating machine has to be air-locked in order to avoid losses in volume and heat of process air.

Front and side access door is air-tightened by inflatable seal mechanism resulting maximum dry efficiency.



Spray Gun Swivel Arm

The spray guns together with the compressed air lines are smoothly taken out of the coating pan through a swivel arm block for easy maintenance and cleaning.



Gun Fan Air Control

Air pressure in coating spray nozzle is critical parameter for coating quality.

Pattern and Atomizing air setting is digital technology not manual so the current value display and adjustment are available on the HMI screen.



Spray Gun

The gun sprays the solution liquid on to the tablet bed. One line for the solution liquid and two lines for compressed air [Cylinder air & atomizing air] are connecting to the gun.

Optional Items



Heat Exchanger
Inlet process air is heated inside this unit prior to introduction to the coating pan.



Dehumidifier
If the cooling & heating coil is required for the inlet air supply, then it can be an option instead of Standard Heat Exchanger at Coating Machine.



Hepa Filter
Hepa filter is equipped on the heat exchanger or dust collector, and the air is filtered to 99.97% [EN 1882 H13].



Dust Collector - Cone Type
The cone type dust collector is designed for high performance. It provides easy maintenance and reduces cleaning frequency with a large and convenient dust collection bin.



Dust Collector
Dust is extracted from the coater via this device which makes for a clean production environment.



WIP Rack
WIP [Wash-In-Place] system automatically cleans each part of the coater.



Liquid Tank [Solution Tank] Air Motor Agitator Impeller
The[Liquid] solution is contained in this tank and transferred to the spray guns by the pump.



Coating Pan Rack
The coating pan rack is for PC-L series. This is for storage of coating pans and plenums.

Optional Items



Auto Damper
A compressed air controlled rotary actuator precisely regulates the airflow to the coater by modulation of the flap position.



Liquid Measurement System
At the laboratory model, the liquid measurement is available with electronic scale option. For the production model, Film Coating, Auto mode, estimated Total Liquid Amount can be checked after the cycle.



Pump Rack
The Pump Rack with Peristaltic Pump supplies the liquid from the Solution Tank to the spray guns in the coating machine. Servo motor control for the pump and it delivers liquid with sufficient force even at low speed.



Coating Pan Change Device
For the pan change at PC-C 620 Series, coating pan change device is required, same quantity of the coating pan.



Multi-Head Pump Rack
Each spray gun is equipped with an individual pump head to prevent minor flow variations and ensure precise and stable liquid supply. Up to 4-5 pump heads can be configured for uniform solution distribution.



High Shear Mixer

The Frontiers of New Technology!



04



* Specifications are subject to change without notice.

PM-C	70
PM-1005~1060	72
PM-1100~1600	73
Optional Items	78

LABORATORY
PM-C

With its interchangeable bowl system, the PTK PM-C series is the best fit for laboratory use, with excellent performance and design.

- Vessel Change
- Agitator Impeller
- Chopper Impeller
- HMI



Model		PM-C001	PM-C003	PM-C005
Vessel Volume	[ℓ]	2.4	5.7	10.1
Working Volume	[ℓ]	0.65 - 1.1	1.7 - 3	3.8 - 6.6
Output Capacity *	[kg]	0.45 - 0.77	1.2 - 2.1	2.7 - 4.6
Agitator Drive Motor	[kW]	0.75		
Agitator Max, Speed	[rpm]	300		
Chopper Drive Motor	[kW]	0.75		
Chopper Max, Speed	[rpm]	3,000		
Dimension	[mm]	1,190 x 555 x 1,450		
Weight	[kg]	550		
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz		
Control Power		24Vdc		

* Specific Gravity 0.7

In-house Program with 21 CFR Part 11 Compliance

PTK's laboratory equipment is equipped with its own stable program. In addition, it is possible to create a report at the global standard level, and it is optimized for data collection by saving the values of important parameters in real-time.



21 CFR Part 11 Compliance
PTK supports the highest level of electronic data and management by establishing an in-house program that complies with 21 CFR PART 11. 'Electronic Records' Global standard level report is automatically generated, and unlimited storage is possible. 'Electronic Signatures' Unlimited permission groups can be set, and electronic signatures for each worker are supported. Audit Trail MDB is stored and managed so that all data can be traced.



Report
PTK's report summarizes all management items stored in the database and saves them in report format. Monitoring values of all parameters are generated in real-time as graphs for intuitive confirmation, and causes and effects can be checked through graph patterns, which enables trend analysis. In addition, it is possible to collect reliable critical process parameters, so it actively responds to QbD.



Parameter Graph
The user-friendly touch screen display and the well-organized production parameter monitoring screen will provide a most satisfactory working environment. With the manual cycle graph, users can check all the real-time parameter graphs in a single table. The figure of every selected parameter shows by percentage for equalized visual graphics, the user can find operational discrepancies easily and deal with in time. This results in the synthetic analysis of cause and effect.

PILOT

PM-1005 ~ PM-1060

The PTK PM-1005, 1015, 1030, 1060 are ideal for laboratory, scale-up or small batch production. User-friendly operation and easy maintenance makes for a highly productive addition to all production environments.

- Agitator Impeller
- Chopper Impeller
- Vessel Lid Opening by Air cylinder
- HMI



Model		PM-1005	PM-1015	PM-1030	PM-1060
Vessel Volume	[ℓ]	14	33	90	130
Working Volume	[ℓ]	3.8 - 6	10.7 - 18	20.6 - 50	44.4 - 75
Output Capacity *	[kg]	2.7 - 4.2	7.5 - 13	14.4 - 35	31 - 53
Agitator Drive Motor	[kW]	0.75	1.5	2.2	2.2
Agitator Max, Speed	[rpm]	300	300	200	150
Chopper Drive Motor	[kW]	0.75	1.5	2.2	2.2
Chopper Max, Speed	[rpm]	3,400			
Dimension	[mm]	1,420 x 555 x 1,470	1,640 x 650 x 1,630	1,250 x 1,240 x 1,810	1,500 x 1,450 x 1,930
Weight	[kg]	650	700	800	950
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz			
Control Power		24Vdc			

* Specific Gravity 0.7

PRODUCTION

PM-1100 ~ PM-1600

For a fast and efficient mixing system, the PTK PM-1100 ~ PM-1600 will exceed all expectations with its user-friendly design.

- Agitator Impeller
- Chopper Impeller
- Vessel Lid Opening by Air cylinder
- HMI



Model		PM-1100	PM-1200	PM-1300	PM-1400	PM-1500	PM-1600
Vessel Volume	[ℓ]	248	440	635	830	1,100	1,270
Working Volume	[ℓ]	85 - 135	142 - 238	172 - 318	254 - 412	330 - 550	390 - 640
Output Capacity *	[kg]	60 - 95	99 - 167	120 - 223	178 - 288	231 - 385	273 - 448
Agitator Drive Motor	[kW]	11	15	22	30	37	37
Agitator Max, Speed	[rpm]	120	110	100	90	80	75
Chopper Drive Motor	[kW]	5.5	7.5	11	11	15	18.5
Chopper Max, Speed	[rpm]	3,400					
Dimension	[mm]	2,450 x 1,550 x 2,500	2,600 x 1,600 x 2,600	2,860 x 1,660 x 2,570	2,850 x 1,900 x 2,570	2,950 x 1,900 x 2,570	3,430 x 2,100 x 2,550
Weight	[kg]	1,950	2,500	2,600	2,750	3,000	3,500
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz					
Control Power		24Vdc					

* Specific Gravity 0.7



Video

Wet Granulation from R&D to Production Capacity



Since wet granulation is a batch process, the amount of material that can be processed in a given batch is dependent on the size of the mixing bowl. Our line of PTK mixer granulators come in a wide range of sizes, from an R&D unit with small bowl with a 1.1 liter working capacity to production granulators with a working bowl volume of approximately 640 liters [approx. 450 Kg capacity.] Product contact parts on all our granulators are constructed of type 316 stainless steel with a mirror finish as standard. Non-product contact exterior surfaces are of brushed type 304 stainless steel.

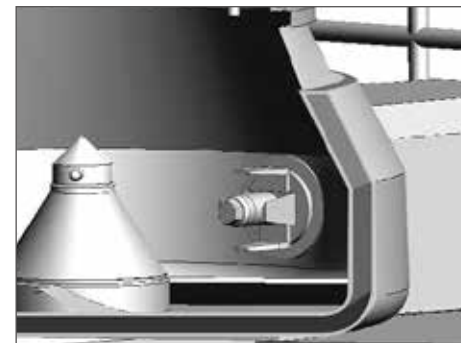


The 'Tulip' Shape of Mixing



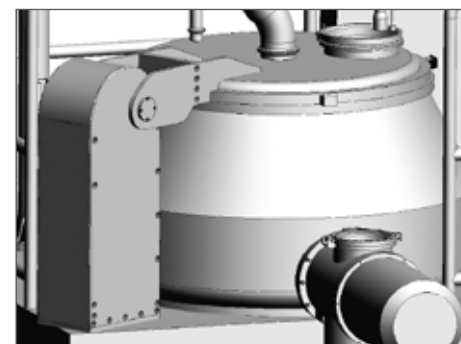
Impeller Lifting Device

Simple and Special One Stop Solution



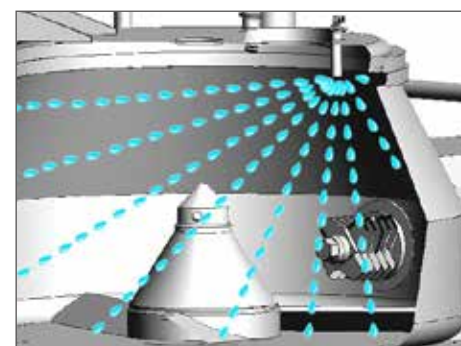
Jacketed Vessel

The vessel is jacketed type. Hot and Cold water supplied to the all cavity to control the vessel temperature



Vessel Lid Opening

The vessel lid is automatically lifted by an air cylinder at the touch of a button. According to room height, slide type or vertical type is available on requests.



Cleaning - WIP

High pressure water is sprayed through purposefully placed nozzles to clean the inside of the vessel. The nozzles can be easily removed and reassembled.

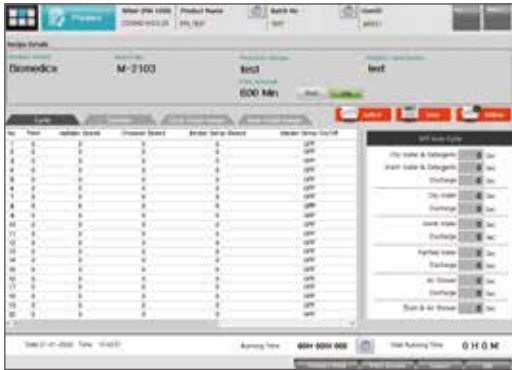
Maximized Customization with In-House Program

The PISCon of the ConPid mixer is designed for easy manipulation of the control system.

The mixing process is automated and configured to the optimal production environment, which guarantees exceptional product stability. The user-friendly touch screen display and the well-organized production parameter monitoring screen will provide a most satisfactory working environment. Batch report and data analysis functions are available with Tablet Coater, High Shear Mixer and Fluid Bed System.



Overview



Recipe

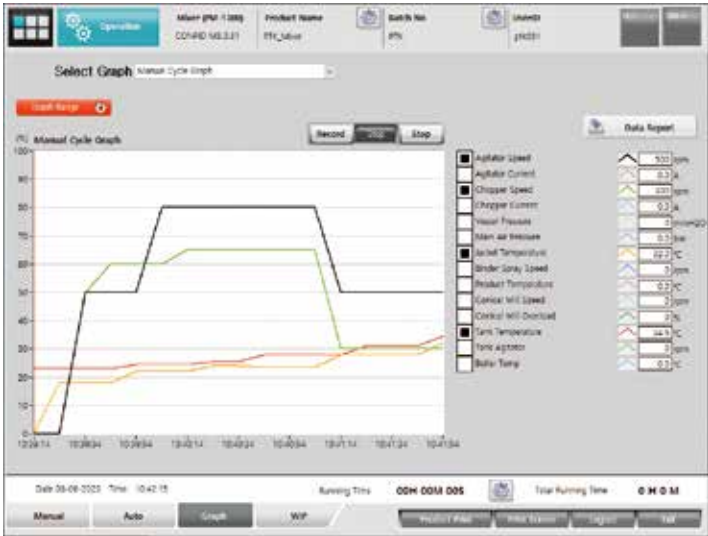


WIP



Batch Report

Data Analysis



With the manual cycle graph, users can check all the real-time parameter graphs in a single table. The figure of every selected parameter shows by percentage for equalized visual graphics, the user can find operational discrepancies easily and deal with in time. This results in synthetic analysis of cause and effect.

This data acquisition affords the ability to learn more about a product's characteristics. If you utilize this data, even heavy processing tasks can be optimized and automated.

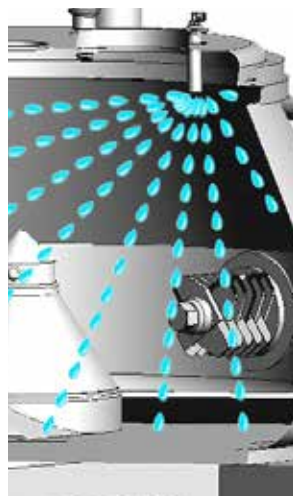


Optional Items



Binder Spray System

The liquid binder is contained separately in its own tank and transferred into the vessel through the binder nozzle. Peristaltic pump with servo motor control.



WIP Rack & Nozzle

High pressure water is sprayed from the WIP Rack to the strategically placed nozzles to clean the inside of the vessel.



Integrated Conical Mill

Built-in type of Conical Mill at the Mixer. Deagglomeration or delumping of wet granules through the integrated conical mill with screen is available. The rotating impeller forces the material outward to a conical mill screen.



Vessel Lid Opening by Air Cylinder

The vessel lid opening & closing operation will be done by the air cylinder, automatically.



Double Jacket

The vessel is jacketed type. Hot and Cold water supplied to the all cavity to control the vessel temperature.



Agitator Impeller Lifting Device

After the WIP process, agitator impeller can be lifted up by the air cylinder to check the cleaning condition of agitator sealing area.

Optional Items



Internal Lamp

The vessel cover has an LED lamp. It helpful when the operator checks the working condition through the window on the vessel Lid. The lamp cover can be disassemble separately for easy cleaning & maintenance.



Vacuum Conveyor

The Vacuum conveyor offers safe and hygienic powder handling. It is used for the transfer of powder, granulated material. Purely pneumatic powered systems with failsafe controls and no tools required for disassembly and cleaning.



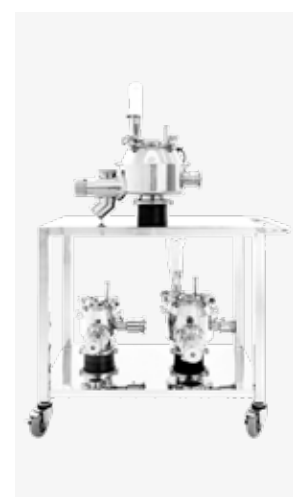
Wire Lifting Device

The heavy agitator can be lifted easily by this system. It is installed at the machine frame with compact design.



Metal Filter

As an Option and customer's requirement, bag filter can be replaced with Stainless Steel filter for easy cleaning and long life span.



Vessel Rack for Additional vessels

At the PM-C model, additional vessels for different capacity can be stored at this vessel rack.



Fluid Bed System

The Frontiers of New Technology!



05



* Specifications are subject to change without notice.

PFB-L	82
PFB-P10	83
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Bin Lifter	97

LABORATORY

PFB-L

The PFB-L series is a lab-scale fluid bed system capable of dry as basic function, granulating with top spray and coating with bottom spray as an option in a single unit. Built-In air handling unit electric heating & filter.

-  Drying
-  Granulating [Top Spray]
-  Coating [Bottom Spray]
-  Metal filter
-  HMI



Video

Model			PFB-L
Container Volume		[ℓ]	18
Working Capacity	Drying *	[kg]	3.5
	Granulating *	[kg]	2.8
	Coating *	[kg]	2
Expansion Chamber Diameter		[mm]	Ø 320
Fan Air Volume		[CMM]	3.3
Fan Differential Pressure		[mmAq]	1,120
Fan Power		[kW]	2.2
Heating Capacity		[kW]	7
Heating Capacity		[kcal/h]	6,020
Compressed Air Pressure		[bar]	6
Dimension, Machine		[mm]	1,400 x 930 x 1,975
Weight, Machine		[kg]	1,000
Power Supply Data			220/380/400/415/440/460Vac, 3Ph, 50/60Hz
Control Power			24Vdc

* Specific Gravity 0.5

PILOT

PFB-P10

As a Pilot model, capable of drying the wet granulation, granulating with top spray and coating process with bottom spray.

-  Drying
-  Granulating [Top Spray]
-  Coating [Bottom Spray]
-  HMI



Model			PFB-P10
Container Volume		[ℓ]	90
Working Capacity	Drying *	[kg]	16
	Granulating *	[kg]	13
	Coating *	[kg]	10
Expansion Chamber Diameter		[mm]	Ø 556
Fan Air Volume		[CMM]	10
Fan Power		[kW]	6.3
Heating Capacity		[kW]	15
Heating Capacity		[kcal/h]	12,900
Compressed Air Pressure		[bar]	6
Dimension, Machine		[mm]	1,740 x 1,282 x 2,095
Weight, Machine		[kg]	1,500
Power Supply Data			220/380/400/415/440/460Vac, 3Ph, 50/60Hz
Control Power			24Vdc

* Specific Gravity 0.5

PRODUCTION

PFB Series

It can be used in the powder processing. simple drying, granulating and particle coating, all fluidized bed processes are available. Wide range of high-performing production technology to meet customer's required batch size.

-  Drying
-  Granulating [Top Spray]
-  Coating [Bottom Spray]
-  HMI

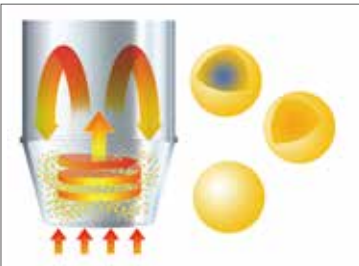


Video

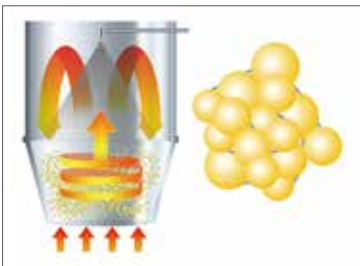
Model		PFB-20	PFB-30	PFB-60	PFB-90	PFB-120	PFB-150	PFB-200	PFB-300
Container Volume [ℓ]		86	125	250	375	500	625	825	1,250
Working Capacity	Drying * [kg]	34	50	100	150	200	250	330	500
	Granulating * [kg]	27	40	80	120	160	200	265	400
	Coating * [kg]	20	30	60	90	120	150	200	300
Fan Unit, Air Volume [CMM]		20	30	50	65	75	85	100	130
Fan Unit, Diff Pressure [mbar]		122.6							
Fan Unit, Power [Kw]		15	15	22	30	30	37.5	45	55
Steam Consumption [kg/h]		50.6	95.6	147.1	166.8	235.4	264.8	353.1	490.5
Steam Pressure [bar]		3							
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz							
Control Power		24Vdc							

* Specific Gravity 0.5

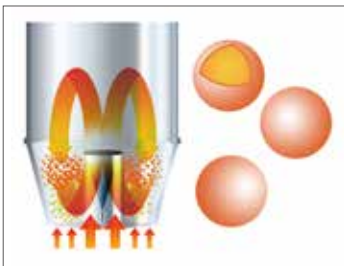
Flexible and Stable
Multi Process



Drying



Granulating [Top Spray]

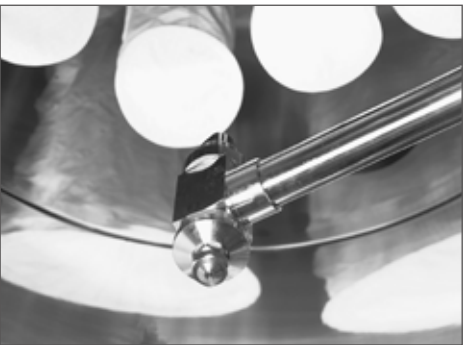


Coating [Bottom Spray]



Drying

Drying is a simple and basic principle. Powder is contained in the lower vessel and heated air is passed through a fine mesh and into the container for the circulation of the targeted powder. Heated air passes through the powder, aerating it to promote rapid drying. The expelled air is then passed through an internal filter, and discharged to the outlet duct by the dust collector. Fluid bed drying guarantees short drying time.



Granulating [Top Spray]

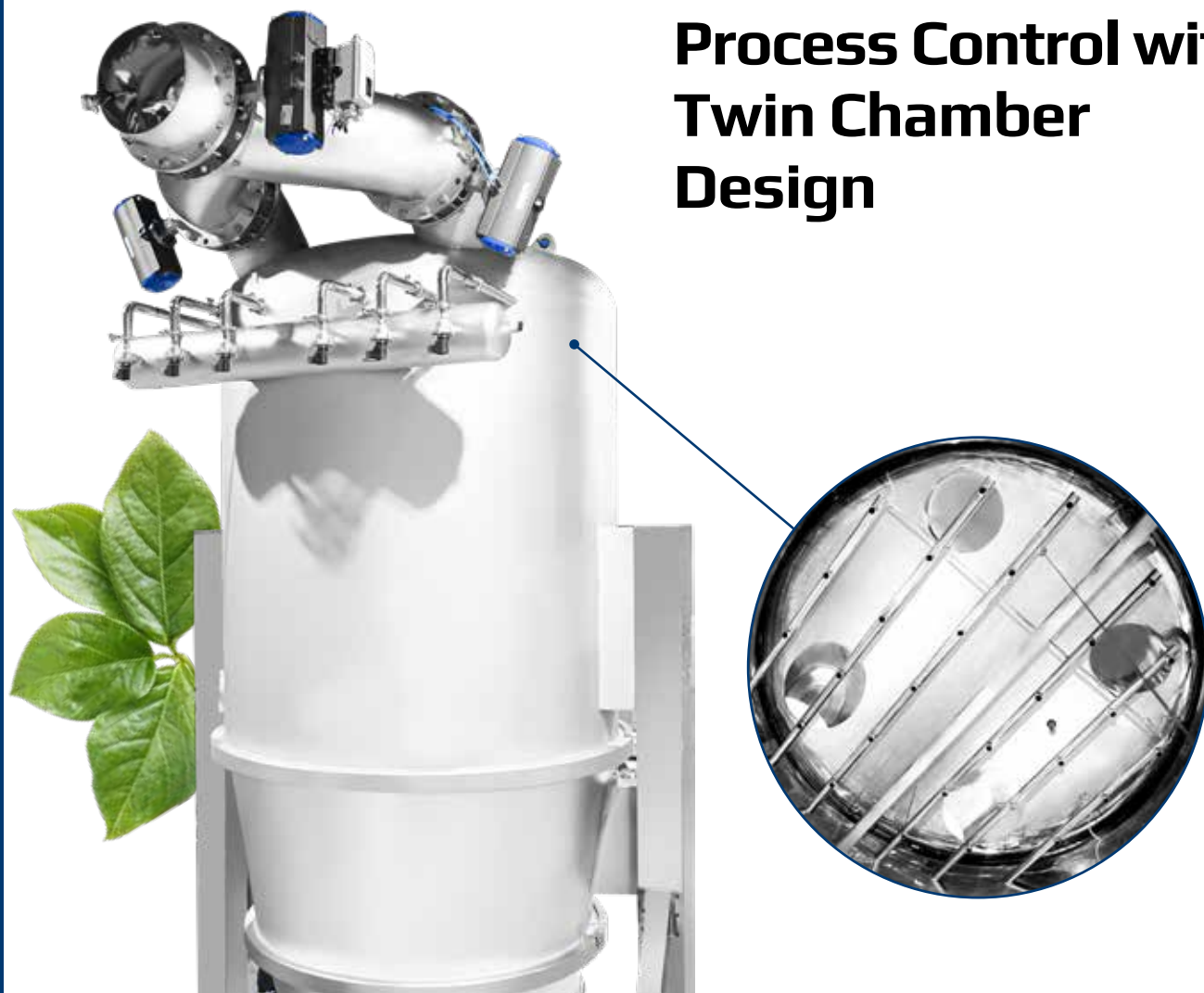
This process makes granulates by spraying various type of liquids onto powder. Spraying is done from the top, the opposite side to fluidization. The multi head nozzle makes it convenient and easy to clean. Due to variance of the volume of the product or the size of the nozzle, the height of the Spray gun is adjustable.



Coating [Bottom Spray]

PFB provides multi process with a single machine. The Drying, Granulating and Coating process are possible with one machine. The coating option includes granulating. Spraying is done from the bottom, With the same method as fluidization. It is best for fine coating. The partition in the container guides the flow of product, from the center to the direction of fluidization. Conversion from top spray to bottom spray systems is possible with product container and spray change, only.

Process Control with Twin Chamber Design



Filtration System with One-Piece Filter



The updated system applies a longer one-piece filter design, increasing the available filtration area and Supporting the dust extraction efficiency. Improved airflow and filtration performance depending on process conditions.

- Reduced drying time
- Stable airflow during operation
- Reduced downtime related to filter clogging

Fluid Bed System with Conidur

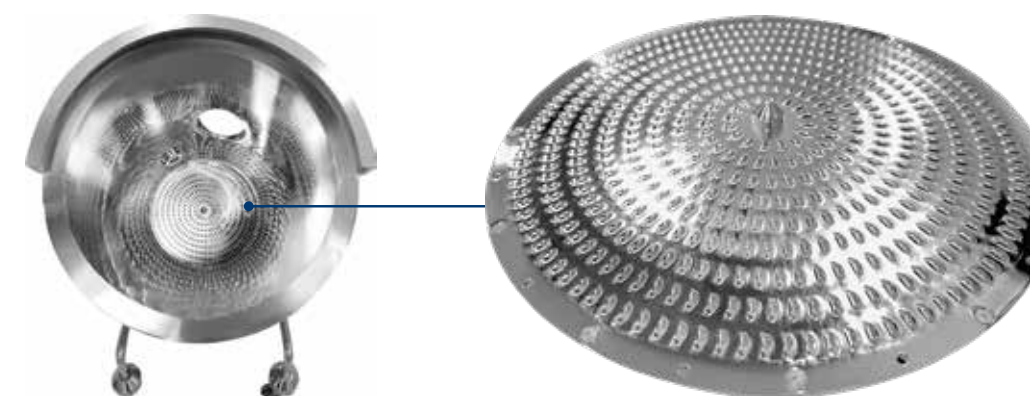
The Conidur structure provides uniform air distribution and stable fluidization throughout the process chamber, minimizing dead zones and channeling. This ensures smooth powder movement and consistent quality consistent quality during the process.

By maintaining stable operating conditions, batch-to-batch variation is reduced and process reproducibility is improved with easy way. In addition, minimal material retention enhances CIP/WIP efficiency, ensuring high hygiene standards in GMP environments.



One of the main design updates is the Twin Chamber structure. The chamber is divided into two sections, allowing airflow to be controlled separately while maintaining stable fluidization during production operation.

- Enables uniform granulation and coating
- Stable airflow distribution in the processing chamber, supporting consistent process conditions
- Improved process stability for production-scale operation, depending on system configuration



Safe Environment and Smart Maintenance



Explosion Safety

PFB series is designed for 12 Bar pressure resistance to protect the user in case of explosion.
The explosion isolation valve [QASV] is also installed on the inlet and outlet ducts. It provides a safe environment even with an explosion flame and pressure.



WIP

WIP nozzles at the filter hosing, expansion chamber and inlet receptacle with WIP Rack provide the automatic cleaning process.
The current WIP cycle and parameters can be monitored on the HMI screen.



Inflatable Seal

Inflatable seal is inflated at filter housing, expansion chamber & etc and expanded when the air is supplied to the seal and provide a perfect closed system.



Filter Maintenance

The fabric bag filter is equipped in the filter housing and the filter is moved up and down by the wiring system with motor.
The differential pressure displayed on the HMI screen so detect the filter blockage.



Proportional Control Device

The proportional valve to rotate butterfly damper installed at the Inlet and Outlet Duct. Operator can adjust & set the open & close rate(%) of in & out damper on the HMI screen. Also, proportional pressure regulator installed at the steam by pass line to maintain the pressure accurately.



Inlet Receptacle

The Inlet Receptacle is designed to air flow in a swirl by the stator. It helps with end product quality and aid's side discharging.

Value Focused Engineering



Loading / Unloading

Powder is loaded by a dust collector and unloaded by an isolated Vacuum conveyor. Finished material is transferred through antistatic hose to a Vacuum conveyor [optional item].

Antistatic hose minimizes sticking inside the hose wall so it can reduce the time to transfer the powder and safely eliminates the risk of electrical static discharge.



Product Temperature Sensor

A sensor for product temperature makes it possible to monitor temperature of products in real time. With high grade stainless steel [316L] and an IP 68 graded sensor guarantees water and dust-free operation.

Additionally, a sanitary clamp makes assembly and disassembly easy.



Sampling

Anytime during the process, user can take sample and monitor the product. With continuous feedback, the user can control all the parameters required by the HMI system.

This reduces the risk of product failure and guarantees high quality.



Side Discharge

- After processing, powder is discharged by a Vacuum conveyor. Once the dust collector makes fluidization of the powder, the Vacuum conveyor carries the powder.
- This closed system gives a dust-free and safe environment even with hazardous products.
- Container tilting system, rotating the handle manually is basic standard feature.



Maximized Customization with In-House Program



In-House Program

With our in-house programmed software, PTK provides maximized customization of the software. This makes for a fast, light and stable program with intuitive interface. Opened protocol means it is possible to connect OEE system which shows real time productivity, availability and quality.



Data Collection

After each batch, the user can get the report of a batch including all main process parameters, events and logs. The batch report is available in both auto and manual mode. With this report, data acquisition [input and output] and analysis it is easily possible to optimize the process for greater efficiency.



Batch Report

The control systems available are dependent upon the customer's specified level of system control and the sophistication level of data collection and analysis to be performed by the control system. An unlimited number of data storage also guarantees maximized traceability.

PRODUCTION CONICAL MILL PCM-20DIM



A cone shaped mill is equipped directly below the point of product discharge, this allows the product to be milled to a consistent size and various grades on discharge by selecting the correct screen type.



Model		PCM-20DIM
Working Max' Capacity*	[kg/h]	2,000
Imeller Max' Speed	[rpm]	1,000
Imeller Drive Motor	[kW]	5
Lifting Motor	[kW]	0.75
Screen Size	[mm]	Ø250 x 138
Screen Hole	Dry Standard	[mm] Ø1.5 (Option : Ø1, Ø2, Ø3)
	Wet Standard	[mm] Square 8x8 (Option : Square 4x4, 5x5, 6x6, 10x10, 12x12)
	Grater	[mm] Ø1 (Option : Ø1.5, Ø2)
Dimension	[mm]	1.145 x 1,360 x 2,308
Weight	[kg]	700
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz
Contol Power		24Vdc

* Depending on the screen hole size and character of the powder, the max capacity can be changed.



LABORATORY CONICAL MILL PCM-L

Laboratory Model conical Mill is also available as stand alone type. Different screen sizes can be provided.

Model		PCM-L
Working Max' Capacity*	[kg/h]	15~120
Imeller Max' Speed	[rpm]	2,400
Imeller Drive Motor	[kW]	1.5
Lifting Motor	[kW]	-
Screen Size	[mm]	Ø130 x 74
Screen Hole	Dry Standard	[mm] Ø1.5 (Option : Ø1, Ø2, Ø3)
	Wet Standard	[mm] Square 8x8 (Option : Square 4x4, 5x5, 6x6, 10x10, 12x12)
	Grater	[mm] Ø1 (Option : Ø1.5, Ø2)
Dimension	[mm]	Ø165 x 261
Weight	[kg]	67
Power Supply Data		220/380/400/415/440/460Vac, 3Ph, 50/60Hz
Contol Power		24Vdc

* Depending on the screen hole size and character of the powder, the max capacity can be changed.



Optional Items



Pump Rack
For Top spray and Bottom spray options, Pump Rack with Peristaltic Pump will be provided. The liquid from the liquid tank to the spray system in the PFB can be delivered with the pump automatically.



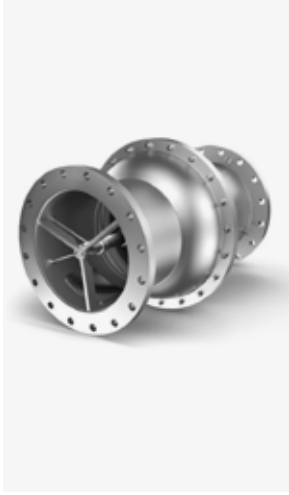
Dehumidifier
As Inlet Air Handling Unit, it is standard feature of the PFB system including heating coil & cooling coil. Pre, Medium and Hepa filter included.



Heat Exchanger
Heat Exchanger is an option for the PFB system instead of Dehumidifier as an economic way. The air heated passing through the heating coil and hot air can be provided to the PFB system.



Dust Collector
The cartridge filters of high filtration efficiency are equipped with regular basis air shaking system activated by compressed air to remove the fine dust on the filter surface.



Explosion Safety
PFB series is designed for 12 Bar pressure resistance to protect the user in case of explosion. The explosion isolation valve (QASV) is also installed on the inlet and outlet ducts. It provides a safe environment even with an explosion flame and pressure.



Closing System
PFB and Mixer can be connected as closing system using automatic inlet & outlet (side discharge) at the PFB machine. The required items for closing, inlet & outlet port, Vacuum conveyor with conical mill & etc are available as options.

Optional Items



Conical Mill with Vacuum Conveyor
Final product can be discharged through the side discharge (outlet port) at the PFB automatically with connection of these system.



WIP Rack
WIP [Wash-In-Place] system automatically cleans each part of the coater.

PRODUCTION BIN BLENDER

PBI-1700 | PBI-1120X

Bin blender Column is helpful for homogenous mixing of dry powders in the food, pharmaceutical and chemical industry. Designed to uniformly blend powders often used in tablet and capsule production. The different capacity of bins can be easily handled, cleaned, assembled and organized.



Video

Model		PBI-1700	PBI-1120X
Capacity	[ℓ]	100 ~ 700	100 ~ 1200
Lifting Speed	[mm/min]	400	400
Blending Speed	[rpm]	2~ 12	2 ~ 7
Lifting Motor Power	[kW]	2.4	2.9
Blending Motor Power	[kW]	2.4	2.9
Electricity	Main	220/380/400/415/440/460Vac, 3Ph, 50/60Hz	
	Control	24Vdc	
	Total Power	[kW]	5
Material		SUS304(SUS316L)	
Bin Type		Round or Square	

LABORATORY BIN BLENDER

PBL-1030

Designed to uniformly blend powders often used in tablet and capsule production.

Model		PBL-1030
Capacity	[ℓ]	2 ~ 30
Blending Speed	[rpm]	1 ~ 25
Blending Motor Power	[kW]	0.75
Electricity	Main	220Vac, 1Ph, 60Hz
	Control	24Vdc
	Total Power	[kW]1
Material		SUS304(SUS316L)
Bin Type		Round or Square



Video

BIN LIFTER

PLT-1800

Different capacity of IBC an be handled at the PTK Bin Lifter. Also the split valve application is available for high containment system with tablet press, capusle filler and tablet coating machine.

Model		PLT-1800
Capacity [ℓ]		Up to 800
Lifting Speed [mm/min]		1,000
Lifting Motor Power [kW]		2.4
Electricity	Main	220/380/400/415/440/460Vac, 3Ph, 50/60Hz
	Control	24Vdc
	Total Power [kW]	2.5
Material		SUS304(SUS316L)
Bin Type		Round or Square





High Containment

The Frontiers of New Technology!



* Specifications are subject to change without notice.

PR - 1500H	100
PF - 1000H	102
Tablet Coater	104
Granulation Line	106

HIGH CONTAINMENT

PR-1500H

The turret (tableting zone) is contained rather than the entire machine in order to use a smaller footprint and volume, not like other bulky containment solutions.
The removable turret can be cleaned easily both on and off the machine.



Deduster Combined with Metal Detector

Bln-Lifter

Tooling		D	B	BB	BBS
Number of Stations		20	24	29	31
Max. Main Compression Force	[kN]	100	80	80	80
Max. Pre Compression Force	[kN]	100	80	80	80
Max. Output	[tabs/h]	120,000	172,800	208,800	223,200
Max. Filling Depth	[mm]	18			
Max. Tablet Thickness	[mm]	8.5			
Upper Punch Insertion	[mm]	1 - 4			
Pitch Circle Diameter	[mm]	285			
Turret Speed	[rpm]	5 - 100	5 - 120	5 - 120	5 - 120
Max. Tablet Diameter (round)	[mm]	25	16	13	11
Max. Tablet Diameter (shape)	[mm]	25	19	14	12
Dimension, Machine	[mm]	1,148 x 1,082 x 1,904			
Dimension, Controls	[mm]	650 x 600 x 1,502			
Weight, Machine	[kg]	2,500			
Weight, Controls	[kg]	120			



WIP Nozzle



Contained Turret with WIP



Turret Table



Video

HIGH CONTAINMENT

PF-1000H

The Capsule and material are fed into the dosing part of the machine without risk of leakage.
A glove port is installed for maintenance and utilizing the cleaning nozzle after WIP sequence.



Video



Capsule Loader

Polisher Combined with Metal Detector

Bin-Lifter

Number of Holes / Segment		1	3	7
Max. Speed, Powder	[rpm]	135		
Max. Speed, Pellet	[rpm]	80		
Max. Output, Powder	[caps/h]	8,100	24,300	56,700
Max. Output, Pellet	[caps/h]	4,800	14,400	33,600
Power of Main Motor	[kW]	1.5		
Air Consumption	[liter/min]	650		
Compressed Air	[bar]	6		
Dimension, Machine	[mm]	1,240 x 1,190 x 2,045		
Dimension, Controls	[mm]	650 x 600 x 1,502		
Weight, Machine	[kg]	1,800		
Weight, Controls	[kg]	120		
Capsule Size		#000, #00, #0, #1, #2, #3, #4		



Inflatable Seals Guarantee Pressure Tightness of the System.



Glove Port on the door for Maintenance.



Capsule Loader



WIP System with Containment Design.

HIGH CONTAINMENT

Tablet Coater

The chamber structured housing with air-tight sealed front and side access doors. Tablets are fed from an IBC lifter in a contained manner and the final coated tablets are discharged into a low profile IBC.



Video



Bln-Lifter

Control Cabinet with HMI System

Model	PC-C620			PC-820	PC-1020	PC-1320	PC-1520	PC-1520i	PC-1720	PC-1720i	PC-1720t
Drum Diameter [mm]	500	600	700	820	1,050	1,300	1,500	1,500	1,700	1,700	1,700
Drum Opening Diameter [mm]	300			380	420	500	578	578	600	600	600
Drum Working Volume [ℓ]	4.0-7.0	9.0-16.0	17.0-28.0	24-45	57-100	100-180	160-280	200-362	250-440	340-600	450-800
Output Capacity * [kg]	2.8-4.9	6.3-11.2	11.9-19.6	17-32	40-70	70-126	112-196	140-253	175-308	238-420	315-560
Pan Speed [rpm]	0 ~ 20			0 ~ 16							0 ~ 12
Spray Gun	1			2	2	2	3	3	4	4	5
Drive Power [kW]	0.75			2.2	2.2	3.7	3.7	3.7	7.5	7.5	7.5
Weight, Machine [kg]	500	600	650	1,450	1,650	1,850	2,000	2,150	2,250	2,400	3,000
Weight, Controls [kg]	90										
Dimension, Machine [mm]	1,050			1,200	1,460	1,720	2,050	2,050	2,254	2,254	2,254
	x 1,463			x 1,610	x 1,585	x 1,705	x 1,818	x 2,038	x 2,008	x 2,258	x 2,648
Dimension, Controls [mm]	x 1,507			x 1,682	x 1,757	x 2,005	x 2,244	x 2,244	x 2,392	x 2,392	x 2,392
	661 x 677 x 1,278										

* Specific Gravity 0.7



Spinning WIP Nozzles for Coverage of the Outside of the Coating Drum.

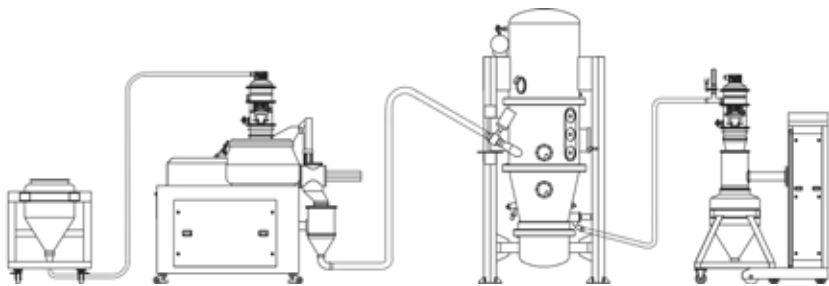


Inflatable Seals for the Front and Side Doors.

HIGH CONTAINMENT

Granulation Line

High shear Mixer and Fluid Bed System can be connected as high containment system.
Powder loading and Discharge will be done with closing system.



BIN



Conical Mill with Vacuum Conveyor

High Shear Mixer

Model	PM-1030	PM-1060	PM-1100	PM-1200	PM-1300	PM-1400	PM-1500	PM-1600
Vessel Volume [ℓ]	90	130	248	440	635	830	1,100	1,270
Working Volume [ℓ]	20.6 - 50	44.4 - 75	85 - 135	142 - 238	172 - 318	254 - 412	330 - 550	390 - 640
Output Capacity * [kg]	14.4 - 35	31 - 53	60 - 95	99 - 167	120 - 223	178 - 288	231 - 385	273 - 448
Agitator Drive Motor [kW]	2	2	11	15	22	30	37	37
Agitator Max, Speed [rpm]	200	150	120	110	100	90	80	75
Chopper Drive Motor [kW]	2	2	5.5	7.5	11	11	15	18.5
Chopper Max, Speed [rpm]	3,400							
Dimension [mm]	1,250 x 1,240 x 1,810	1,500 x 1,450 x 1,930	2,450 x 1,550 x 2,500	2,600 x 1,600 x 2,600	2,860 x 1,660 x 2,570	2,850 x 1,900 x 2,570	2,950 x 1,900 x 2,570	3,430 x 2,100 x 2,550
Weight [kg]	800	950	1,950	2,500	2,600	2,750	3,000	3,500

* Specific Gravity 0.7

Fluid Bed System

Model	PFB-20	PFB-30	PFB-60	PFB-90	PFB-120	PFB-150	PFB-200	PFB-300
Container Volume [ℓ]	86	125	250	375	500	625	825	1,250
Working Capacity : Drying * [kg]	34	50	100	150	200	250	330	500
Working Capacity : Granulating * [kg]	27	40	80	120	160	200	265	400
Working Capacity : Coating * [kg]	20	30	60	90	120	150	200	300
Fan Unit, Air Volume [CMM]	20	30	50	65	75	85	100	130
Fan Unit, Diff Pressure [mbar]	122.6							
Fan Unit, Power [Kw]	15	15	22	30	30	37.5	45	55
Steam Consumption [kg/h]	50.6	95.6	147.1	166.8	235.4	264.8	353.1	490.5h
Steam Pressure [bar]	3							

* Specific Gravity 0.5



Complete Line of
Solid Dose Machinery



All Model Product Catalog

Version | V4.2



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